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# 1. Introduction

This document models connected coffee machines that send telemetry to a central cloud site, receive OTA firmware updates, and produce operational and audit logging evidence. Authored functional architecture remains stable while runtime, code, and verification layers are inferred from infrastructure, source, and test artifacts.

## 1.1. Scope and Assumptions

- Functional Groups and Functional Units are the stable authored design backbone.
- Runtime and code realization are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- Requirement alignment and reference indexes are provided for traceability, not as primary narrative flow.

# 2. How To Read This Document

This document is intentionally split into two layers:

1. Concept layer: functional and view-specific architecture narratives for human understanding.
2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
3. Traceability View: requirement mapping, evidence coverage, and confidence.
4. Reference Appendix: supporting reference IDs and inferred-source links.

## 3. Document Health Snapshot

| View | Authored Status | Authored Status Explanation | Units in Scope | Direct Evidence Coverage | Gap Count | Heuristic Confidence | Heuristic Basis |
|---|---|---|---|---|---|---|---|
| Architecture Intent View (`VIEW-ARCHITECTURE-INTENT`) | stable | Stable authored capability boundaries and responsibility ownership. | 7 | 7 | 0 | high | Defaults authored units to covered; use authoredStatus as the primary publication signal. |
| Communication View (`VIEW-COMMUNICATION`) | in-review | Communication handoffs are mapped; protocol detail remains partially inferred. | 7 | 5 | 2 | medium | Counts a unit as covered when direct inferred runtime evidence exists. |
| Deployment View (`VIEW-DEPLOYMENT`) | in-review | Deployment/runtime mapping is present with evolving control-plane evidence depth. | 7 | 5 | 2 | medium | Counts a unit as covered when direct inferred runtime evidence exists. |
| Security View (`VIEW-SECURITY`) | draft | Core threats are modeled while additional control evidence is being expanded. | 7 | 6 | 1 | high | Counts a unit as covered when explicit authored attack-vector mapping exists. |
| Traceability View (`VIEW-TRACEABILITY`) | in-review | Requirement mappings exist; verification and evidence coverage continue to be hardened. | 7 | 7 | 2 | high | Counts a unit as covered when direct inferred runtime or code evidence exists. |

## 4. Terms and Definitions

| Term | Definition |
|---|---|
| actor (EM-ACTOR) | A person or operational role that interacts with functional units. |
| attack vector (EM-ATTACK-VECTOR) | A technical misuse or attack path that targets functional, referenced, or runtime elements. |

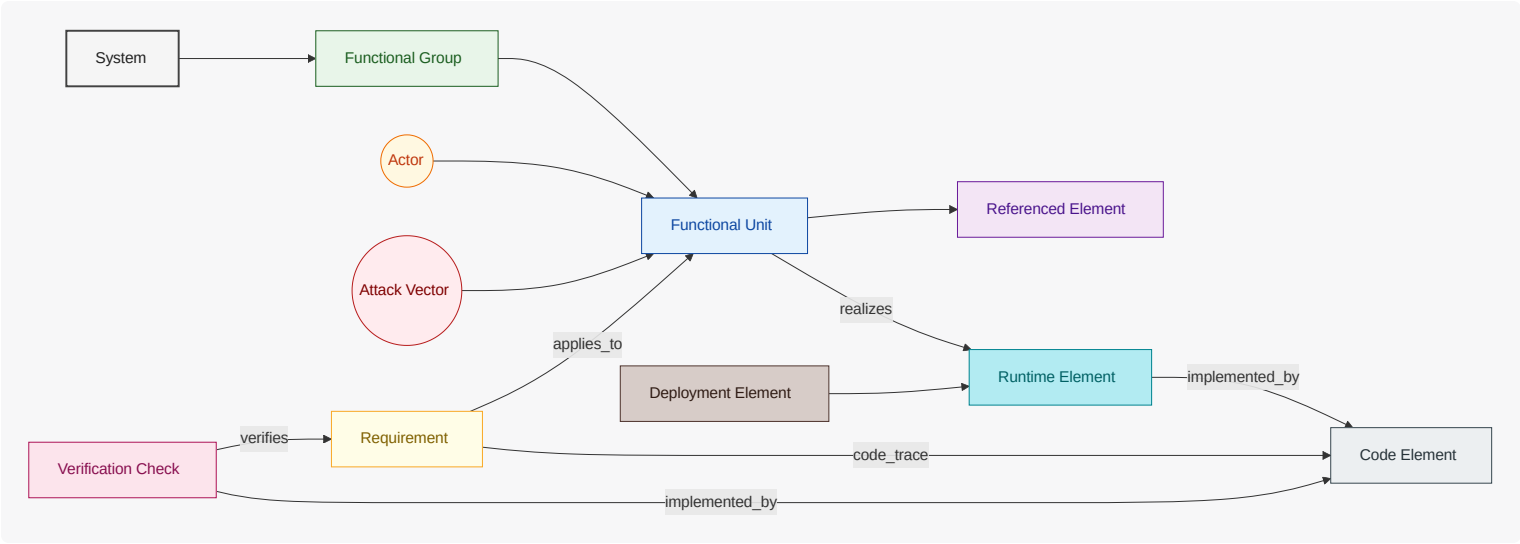
| Term | Definition |
| --- | --- |
| audit log record (DATA-AUDIT-LOG-RECORD) | Immutable log record for security and compliance analysis. |
| audit logging is enabled (FEAT-AUDIT-LOGGING) | Feature flag enabling audit-grade log persistence. Aliases:audit enabled |
| authored mapping (EM-AUTHORED-MAPPING) | An explicit relationship declared in architecture inputs between authored or referenced elements. |
| barista (ACT-BARISTA) | Operates coffee machines and observes local machine health. Aliases:store operator |
| brew cycle completed event is received (EVT-BREW-CYCLE-COMPLETED) | Event raised when a coffee machine completes a brew cycle. Aliases:brew completed |
| brew telemetry (DATA-BREW-TELEMETRY) | Machine telemetry including brew duration, temperature, and status metadata. |
| cloud control (FG-CLOUD-CONTROL) | Capability area for fleet ingestion, campaign orchestration, and cloud reporting. Aliases:central cloud site |
| cloud logging service (REF-CLOUD-LOGGING-SERVICE) | Managed log and metrics service used for fleet observability. Aliases:logging service |
| cloud runtime operations (FU-CLOUD-RUNTIME-OPERATIONS) | Operates cloud runtime deployment and scaling controls. Aliases:runtime operations |
| cloud runtime sdk (REF-CLOUD-RUNTIME-SDK) | SDK used by cloud units for runtime integrations. Aliases:cloud sdk |
| code element (EM-CODE-ELEMENT) | An inferred code structure or ownership element discovered from source trees and build metadata. |
| coffee machine fleet operation is active (STATE-FLEET-OPERATION-ACTIVE) | State where machines and central cloud are actively serving operations. Aliases:fleet operation active |
| device identity and secrets (FU-DEVICE-IDENTITY-SECRETS) | Manages device identity material and secure configuration. Aliases:device secrets |
| firmware bundle store (REF-FIRMWARE-BUNDLE-STORE) | Object store containing signed OTA firmware bundles. Aliases:firmware store |
| firmware signature is valid (COND-FIRMWARE-SIGNATURE-VALID) | Condition indicating firmware signature verification succeeded. Aliases:signature valid |
| firmware signer service (REF-FIRMWARE-SIGNER-SERVICE) | Service that signs firmware artifacts before OTA rollout. Aliases:signer service |

| Term | Definition |
| --- | --- |
| firmware validation failed event is received (EVT-FIRMWARE-VALIDATION-FAILED) | Event raised when signature/hash validation fails on firmware. Aliases:validation failed |
| fleet ingestion api (FU-FLEET-INGESTION-API) | Ingests telemetry and machine status from edge devices. Aliases:ingestion api |
| fleet observability reporting (FU-FLEET-OBSERVABILITY-REPORTING) | Aggregates telemetry and update outcomes into operational and audit signals. Aliases:observability reporting |
| fleet operator (ACT-FLEET-OPERATOR) | Owns OTA rollout campaigns and fleet-level reliability. Aliases:fleet admin |
| functional group (EM-FUNCTIONAL-GROUP) | A major authored capability area that groups related functional units. |
| functional unit (EM-FUNCTIONAL-UNIT) | An authored working unit inside a functional group that owns specific behavior. |
| inferred mapping (EM-INFERRED-MAPPING) | A discovered relationship that links inferred runtime/code elements upward to authored design. |
| iot ingest endpoint (REF-IOT-INGEST-ENDPOINT) | Cloud endpoint for receiving telemetry and status from devices. Aliases:iot endpoint |
| log tampering (AV-LOG-TAMPERING) | Attempts to suppress or alter security/operational log records. Aliases:audit tampering |
| machine edge (FG-MACHINE-EDGE) | Capability area for device-local telemetry and OTA execution behavior. Aliases:coffee machine edge |
| machine telemetry collection (FU-MACHINE-TELEMETRY-COLLECTION) | Collects brew/device telemetry and publishes cloud-bound payloads. Aliases:telemetry collection |
| malicious firmware bundle (AV-MALICIOUS-FIRMWARE-BUNDLE) | Tampered firmware payload attempts to execute on devices. Aliases:firmware tampering |
| mqtt device sdk (REF-MQTT-DEVICE-SDK) | SDK used by edge components for telemetry and command transport. Aliases:mqtt sdk |
| normal mode is active (MODE-NORMAL) | Default operation mode for telemetry and OTA behavior. Aliases:normal mode |
| ota campaign is enabled (FEAT-OTA-CAMPAIGN) | Feature flag enabling centralized OTA rollout behavior. Aliases:ota enabled |

| Term | Definition |
| --- | --- |
| ota campaign policy (DATA-OTA-CAMPAIGN-POLICY) | Policy describing rollout cohorts, windows, and rollback criteria. |
| ota campaign scheduled event is received (EVT-OTA-CAMPAIGN-SCHEDULED) | Event raised when a fleet OTA campaign is scheduled. Aliases:campaign scheduled |
| ota status (DATA-OTA-STATUS) | Device-level status for firmware download, apply, and rollback outcomes. |
| ota status reported event is received (EVT-OTA-STATUS-REPORTED) | Event raised when device reports OTA apply status. Aliases:ota status reported |
| ota update agent (FU-OTA-UPDATE-AGENT) | Downloads, validates, and applies signed firmware updates on machines. Aliases:device update agent |
| platform operations (FG-PLATFORM-OPERATIONS) | Capability area for runtime operations and device identity/secrets lifecycle. Aliases:fleet platform operations |
| platform operator (ACT-PLATFORM-OPERATOR) | Owns runtime deployment and operational controls. Aliases:platform ops |
| referenced element (EM-REFERENCED-ELEMENT) | An architecture-relevant external, platform, or third-party dependency represented by role. |
| runtime element (EM-RUNTIME-ELEMENT) | An inferred runtime realization element discovered from infrastructure and deployment sources. |
| security analyst (ACT-SECURITY-ANALYST) | Reviews identity, firmware integrity, and logging controls. Aliases:security reviewer |
| spoofed device identity (AV-SPOOFED-DEVICE-IDENTITY) | Unauthorized actor impersonates a machine identity. Aliases:device spoofing |
| telemetry ingested event is received (EVT-TELEMETRY-INGESTED) | Event raised when cloud ingestion accepts telemetry. Aliases:telemetry ingested |
| telemetry replay abuse (AV-TELEMETRY-REPLAY-ABUSE) | Replayed telemetry floods ingestion and distorts reporting. Aliases:telemetry replay |
| update campaign orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION) | Selects target machine groups and coordinates OTA rollout phases. Aliases:campaign orchestration |
| upward linking (EM-UPWARD-LINKING) | Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs. |

## 4.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.




# 5. Architecture Intent View

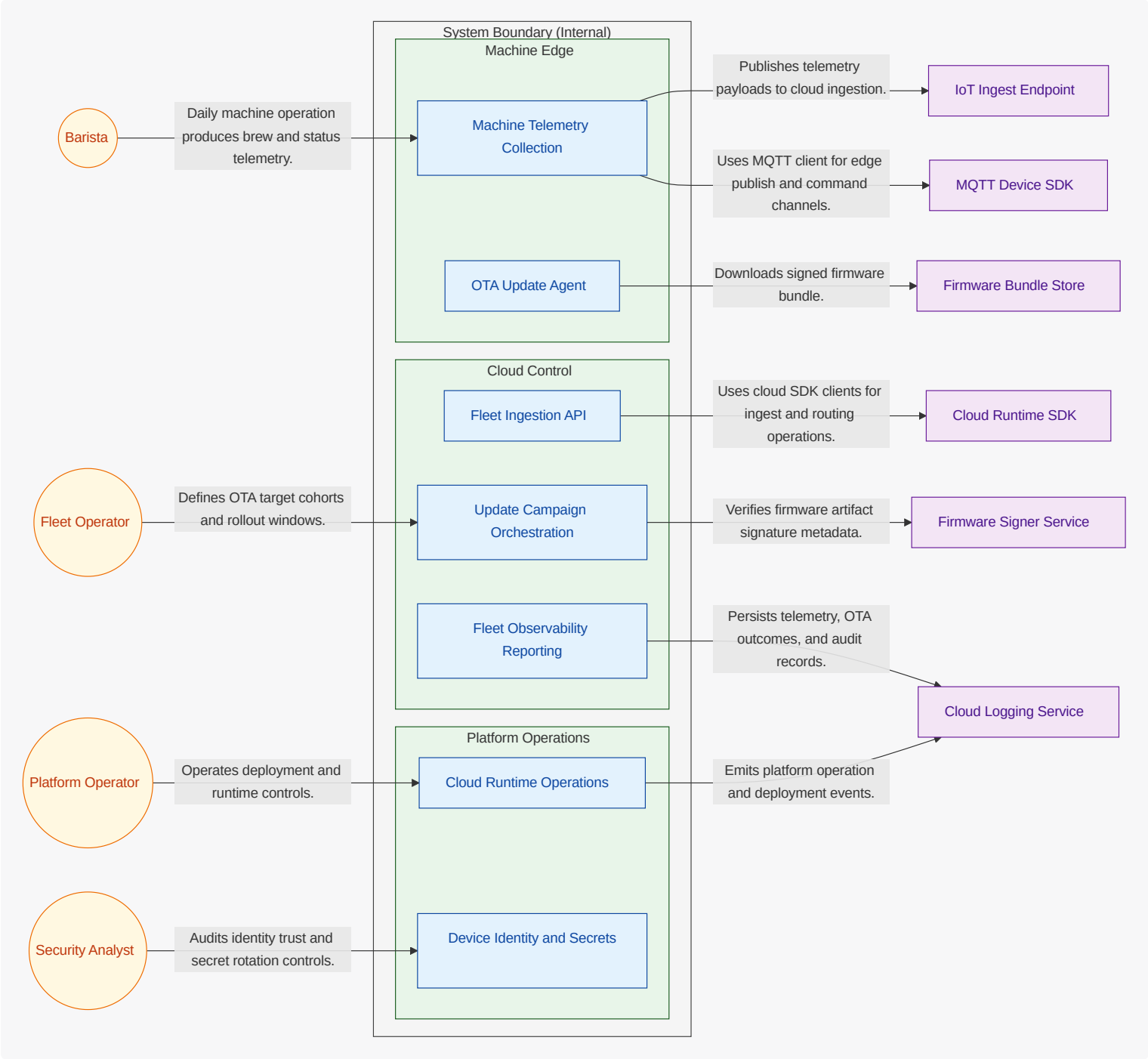
Show the stable authored architecture intent: capability areas, unit ownership boundaries, and intentional responsibility split.

## 5.1. What This View Answers

- What are the main capability areas?
- What are the stable units of responsibility and what does each unit own?
- How are responsibilities intentionally split?

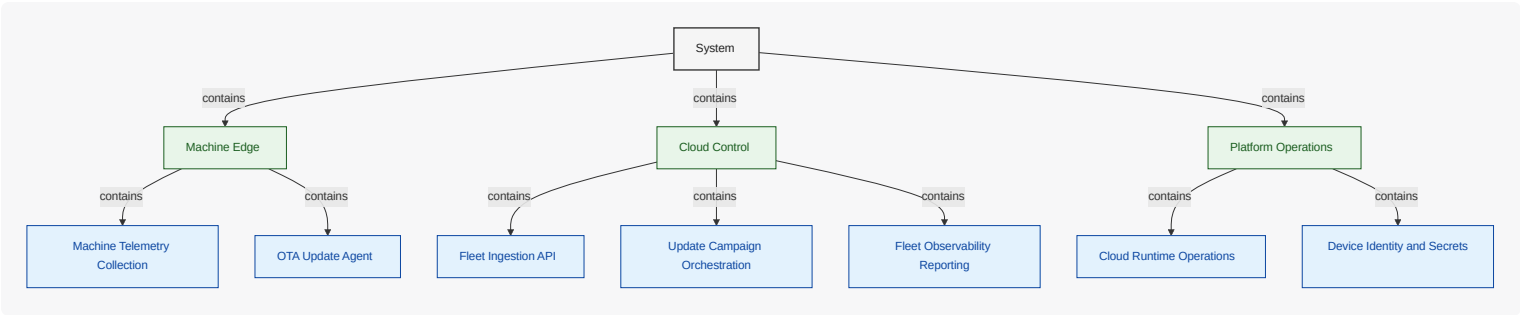
## 5.2. System Boundary Diagram

What this diagram shows: explicit internal vs external scope. Functional units are inside the `System Boundary (Internal)` container, while actors and referenced dependencies are shown outside the boundary.




## 5.3. System Decomposition Diagram

What this diagram shows: stable capability decomposition from functional groups into functional units.



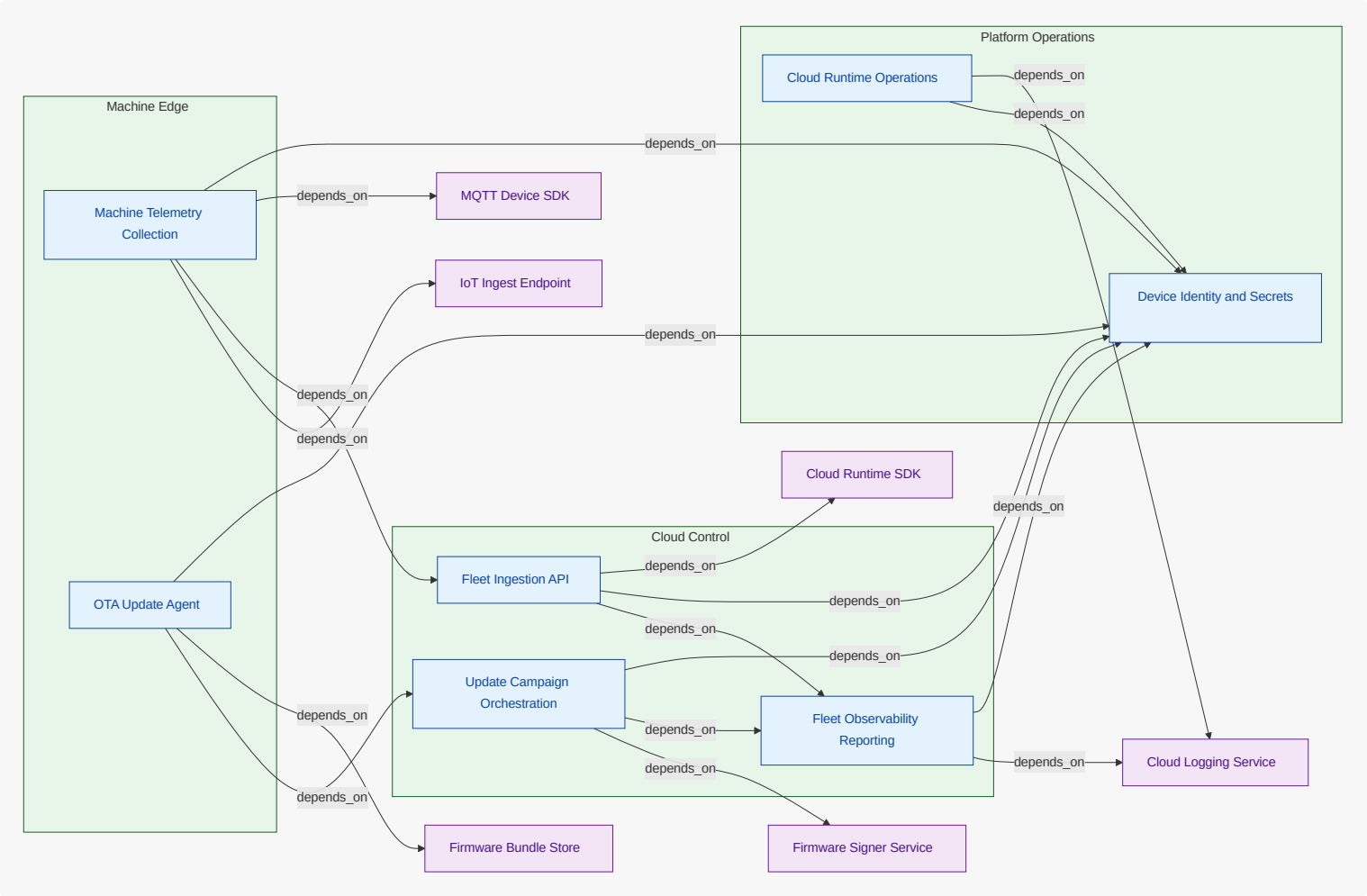
## 5.4. Functional Group Manhattan Matrix

What this diagram shows: a table-like matrix where functional groups are the bottom row (columns) and functional units are arranged in rows above their owning group.

| | | |
|---|---|---|
| | Fleet Ingestion API | |
| Machine Telemetry Collection | Fleet Observability Reporting | Cloud Runtime Operations |
| OTA Update Agent | Update Campaign Orchestration | Device Identity and Secrets |
| Machine Edge | Cloud Control | Platform Operations |

## 5.5. Functional Unit Dependency Diagram

What this diagram shows: dependency paths between functional units (grouped in Functional Group containers) and key referenced elements. Arrow meaning: `A -→|depends_on| B` means unit `A` requires behavior or data from `B` to fulfill its responsibility.




## 5.6. Functional Groups and Units (View Scoped)

### 5.6.1. Machine Edge (FG-MACHINE-EDGE)

Device-local telemetry and OTA execution responsibilities.

Machine Edge owns behavior that executes directly on coffee machines. It gathers telemetry, applies OTA updates, and reports local status to cloud boundaries.

### Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Collects brew/device telemetry and transmits cloud-bound telemetry payloads.

#### Owned Responsibility

functional responsibility for machine telemetry collection; includes decision and orchestration flow to Device Identity and Secrets and decision and orchestration flow to Fleet Ingestion API

#### Key Collaborators

Device Identity and Secrets, Fleet Ingestion API, IoT Ingest Endpoint, MQTT Device SDK

#### Linked Requirements

REQ-COF-001, REQ-COF-006

### OTA Update Agent (FU-OTA-UPDATE-AGENT)

Downloads signed firmware bundles, validates integrity, applies updates, and reports status.

#### Owned Responsibility

functional responsibility for ota update agent; includes decision and orchestration flow to Device Identity and Secrets and decision and orchestration flow to Firmware Bundle Store

#### Key Collaborators

Device Identity and Secrets, Firmware Bundle Store, Update Campaign Orchestration

#### Linked Requirements

REQ-COF-003, REQ-COF-004, REQ-COF-005

## 5.6.2. Cloud Control (FG-CLOUD-CONTROL)

Central cloud ingestion, campaign control, and fleet reporting.

Cloud Control owns central APIs and coordination logic for fleet telemetry and OTA rollouts. It provides global fleet visibility and deterministic rollout decisions.

### Fleet Ingestion API (FU-FLEET-INGESTION-API)

Receives telemetry and machine status, applies validation, and emits ingestion events.

#### Owned Responsibility

functional responsibility for fleet ingestion api; includes decision and orchestration flow to Cloud Runtime SDK and decision and orchestration flow to Device Identity and Secrets

#### Key Collaborators

Cloud Runtime SDK, Device Identity and Secrets, Fleet Observability Reporting

#### Linked Requirements

REQ-COF-001, REQ-COF-002, REQ-COF-006, REQ-COF-008

### Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Aggregates telemetry and OTA outcomes into dashboards, alerts, and audit records.

#### Owned Responsibility

functional responsibility for fleet observability reporting; includes decision and orchestration flow to Cloud Logging Service and decision and orchestration flow to Device Identity and Secrets

#### Key Collaborators

Cloud Logging Service, Device Identity and Secrets

#### Linked Requirements

REQ-COF-002, REQ-COF-005, REQ-COF-007, REQ-COF-008

### Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Plans OTA rollout cohorts, drives update commands, and handles rollback flow decisions.

#### Owned Responsibility

functional responsibility for update campaign orchestration; includes decision and orchestration flow to Device Identity and Secrets and decision and orchestration flow to Firmware Signer Service

#### Key Collaborators

Device Identity and Secrets, Firmware Signer Service, Fleet Observability Reporting

#### Linked Requirements

REQ-COF-003, REQ-COF-005

## 5.6.3. Platform Operations (FG-PLATFORM-OPERATIONS)

Cloud runtime operations plus identity and secret-management controls.

Platform Operations owns deployability, runtime guardrails, and secure identity/config lifecycle. It ensures cloud and device trust boundaries remain enforceable over time.

### Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Owns runtime deployment lifecycle, scaling controls, and runtime operational safety.

#### Owned Responsibility

functional responsibility for cloud runtime operations; includes decision and orchestration flow to Cloud Logging Service and decision and orchestration flow to Device Identity and Secrets

#### Key Collaborators

Cloud Logging Service, Device Identity and Secrets

#### Linked Requirements

REQ-COF-008

### Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Manages device credentials, cloud tokens, and secure configuration used by edge/cloud units.

#### Owned Responsibility

functional responsibility for device identity and secrets

## Key Collaborators

none

## Linked Requirements

REQ-COF-004, REQ-COF-007

# 6. Communication View

Show who communicates with whom across key request paths, including interface edges and dependency handoffs.

## 6.1. What This View Answers

- Who talks to whom and over which interfaces?
- Which edges are synchronous, asynchronous, event-driven, or callback-oriented?
- Which communication paths are critical to the main request flow?

## 6.2. Coverage Gaps

Coverage summary: 5/7 units have direct evidence coverage in this view.

- Cloud Runtime Operations has no direct inferred runtime/deployment evidence yet.
- Device Identity and Secrets has no direct inferred runtime/deployment evidence yet.

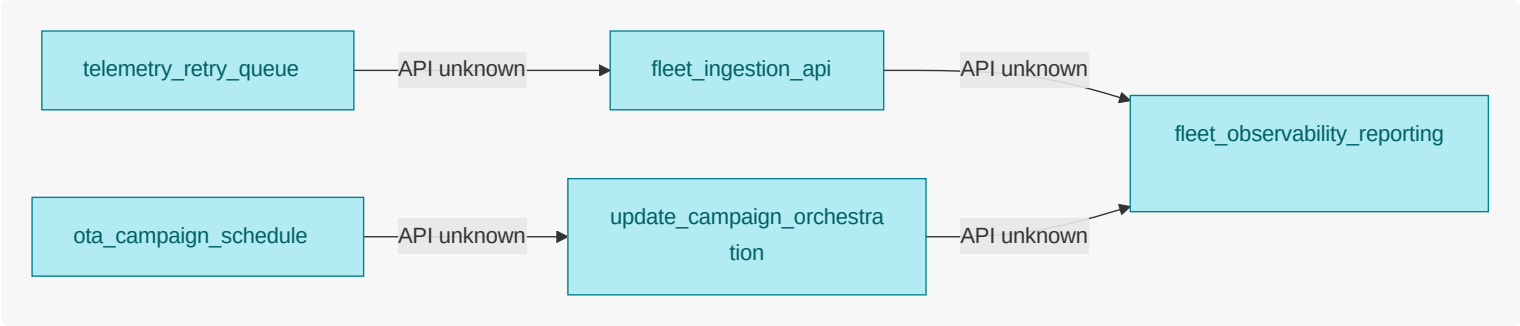
## 6.3. Recommended Next Evidence Additions

- Add explicit communication metadata (protocol, sync/async/event/callback) to authored mappings where currently implicit.
- Expose service protocol/port metadata in Helm or manifest values to enrich interface-edge inference.
- Capture missing communicating runtime units by adding discoverable workload artifacts under scanned inference roots.

## 6.4. Communication Path Graph

What this diagram shows: inferred unit-to-unit request/call flow, including available interface/port labels.

> **NOTE** This graph is inferred from available deployment/runtime artifacts and may be partial. Missing edges or interface details indicate missing derived communication evidence, not necessarily missing behavior.




## 6.5. Functional Groups and Units (View Scoped)

### 6.5.1. Machine Edge (FG-MACHINE-EDGE)

Machine Edge sends telemetry upstream, receives OTA commands, and reports apply/rollback status.

#### Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Sends telemetry over authenticated channels and retries during endpoint outage windows.

#### Inbound Interfaces

| Interface | Description |
|---|---|
| Barista | Daily machine operation produces brew and status telemetry. |

#### Outbound Interfaces

| Interface | Description |
|---|---|
| Device Identity and Secrets | Uses machine identity material for authenticated publish. |
| Fleet Ingestion API | Sends normalized telemetry records to ingestion API. |
| IoT Ingest Endpoint | Publishes telemetry payloads to cloud ingestion. |
| MQTT Device SDK | Uses MQTT client for edge publish and command channels. |

#### Communication Evidence

runtime: telemetry_retry_queue | code modules: machine_telemetry

### OTA Update Agent (FU-OTA-UPDATE-AGENT)

Receives campaign commands and reports update/rollback status back to cloud control.

#### Inbound Interfaces

none

#### Outbound Interfaces

| Interface | Description |
|---|---|
| Device Identity and Secrets | Uses trust anchors for firmware and endpoint authentication. |
| Firmware Bundle Store | Downloads signed firmware bundle. |
| Update Campaign Orchestration | Receives OTA rollout commands and policy constraints. |

#### Communication Evidence

runtime: ota_campaign_schedule, ota_campaign_target | code modules: ota_update_agent

## 6.5.2. Cloud Control (FG-CLOUD-CONTROL)

Cloud Control coordinates telemetry intake, campaign commands, and observability event fanout.

### Fleet Ingestion API (FU-FLEET-INGESTION-API)

Accepts machine telemetry and forwards normalized records to observability reporting.

#### Inbound Interfaces

| Interface | Description |
| --- | --- |
| Machine Telemetry Collection | Sends normalized telemetry records to ingestion API. |

#### Outbound Interfaces

| Interface | Description |
| --- | --- |
| Cloud Runtime SDK | Uses cloud SDK clients for ingest and routing operations. |
| Device Identity and Secrets | Validates device identity and token claims. |
| Fleet Observability Reporting | Forwards validated telemetry and machine status events. |

#### Communication Evidence

runtime: fleet_event_topic, fleet_ingestion_api | code modules: fleet_ingestion_api

### Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Consumes telemetry and OTA outcomes and emits dashboards/alerts/log events.

#### Inbound Interfaces

| Interface | Description |
| --- | --- |
| Fleet Ingestion API | Forwards validated telemetry and machine status events. |
| Update Campaign Orchestration | Emits OTA rollout and rollback events for fleet visibility. |

#### Outbound Interfaces

| Interface | Description |
| --- | --- |
| Cloud Logging Service | Persists telemetry, OTA outcomes, and audit records. |
| Device Identity and Secrets | Uses controlled credentials for audit-log writes. |

#### Communication Evidence

runtime: fleet_observability_reporting | code modules: fleet_observability_reporting

### Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Sends OTA commands to update agents and ingests status feedback for campaign control.

#### Inbound Interfaces

| Interface | Description |
| --- | --- |
| Fleet Operator | Defines OTA target cohorts and rollout windows. |

| Interface | Description |
| --- | --- |
| OTA Update Agent | Receives OTA rollout commands and policy constraints. |

#### Outbound Interfaces

| Interface | Description |
| --- | --- |
| Device Identity and Secrets | Uses trusted control-plane credentials. |
| Firmware Signer Service | Verifies firmware artifact signature metadata. |
| Fleet Observability Reporting | Emits OTA rollout and rollback events for fleet visibility. |

#### Communication Evidence

runtime: update_campaign_orchestration | code modules: update_campaign_orchestrator

### 6.5.3. Platform Operations (FG-PLATFORM-OPERATIONS)

Platform operations coordinates runtime controls and secret distribution contracts with dependent units.

#### Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Coordinates deployment/runtime controls that shape fleet service availability.

#### Inbound Interfaces

| Interface | Description |
| --- | --- |
| Platform Operator | Operates deployment and runtime controls. |

#### Outbound Interfaces

| Interface | Description |
| --- | --- |
| Cloud Logging Service | Emits platform operation and deployment events. |
| Device Identity and Secrets | Requires runtime credentials and encrypted config. |

#### Communication Evidence

code modules: cloud_runtime_ops

#### Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Provides identity validation and secure config interfaces to edge and cloud units.

#### Inbound Interfaces

| Interface | Description |
| --- | --- |
| Cloud Runtime Operations | Requires runtime credentials and encrypted config. |

| Interface | Description |
| --- | --- |
| Fleet Ingestion API | Validates device identity and token claims. |
| Fleet Observability Reporting | Uses controlled credentials for audit-log writes. |
| Machine Telemetry Collection | Uses machine identity material for authenticated publish. |
| OTA Update Agent | Uses trust anchors for firmware and endpoint authentication. |
| Security Analyst | Audits identity trust and secret rotation controls. |
| Update Campaign Orchestration | Uses trusted control-plane credentials. |

### Outbound Interfaces

none

### Communication Evidence

code modules: device_identity_secrets

# 7. Deployment View

Show inferred deployment artifacts and ownership mapping to authored units. This view emphasizes deployment relationships and platform operations.

## 7.1. What This View Answers

- How does authored intent become deployed runtime?
- What control-plane objects manage deployable units?
- Where are platform ownership boundaries and missing ownership signals?

## 7.2. Coverage Gaps

Coverage summary: 5/7 units have direct evidence coverage in this view.

- Cloud Runtime Operations has no direct inferred runtime/deployment evidence yet.
- Device Identity and Secrets has no direct inferred runtime/deployment evidence yet.

## 7.3. Recommended Next Evidence Additions

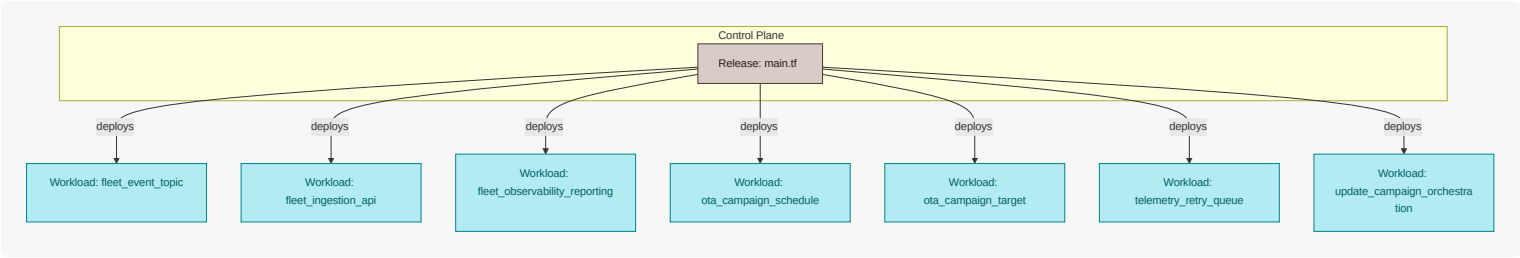
- Add owner metadata to Flux/Kustomization/Helm objects for clearer platform ownership mapping.
- Ensure deployment control artifacts (source, kustomization, release) are all included in scanned directories.
- Add deterministic naming conventions for release and namespace objects to improve cross-artifact linking.

## 7.4. Deployment Artifact Relationship Graph

What this diagram shows: how source/control-plane artifacts connect to releases, namespaces, workloads, and target cluster context.

**NOTE**

This graph is inferred from available deployment artifacts (for example Terraform, Flux, Helm) and may be partial when source metadata is incomplete. Namespace and cluster boundaries are rendered as containers when target/placement relationships are inferred. Node labels include role hints (for example `Source`, `Kustomization`, `Release`, `Workload`, `ns/…`) to distinguish similarly named elements.

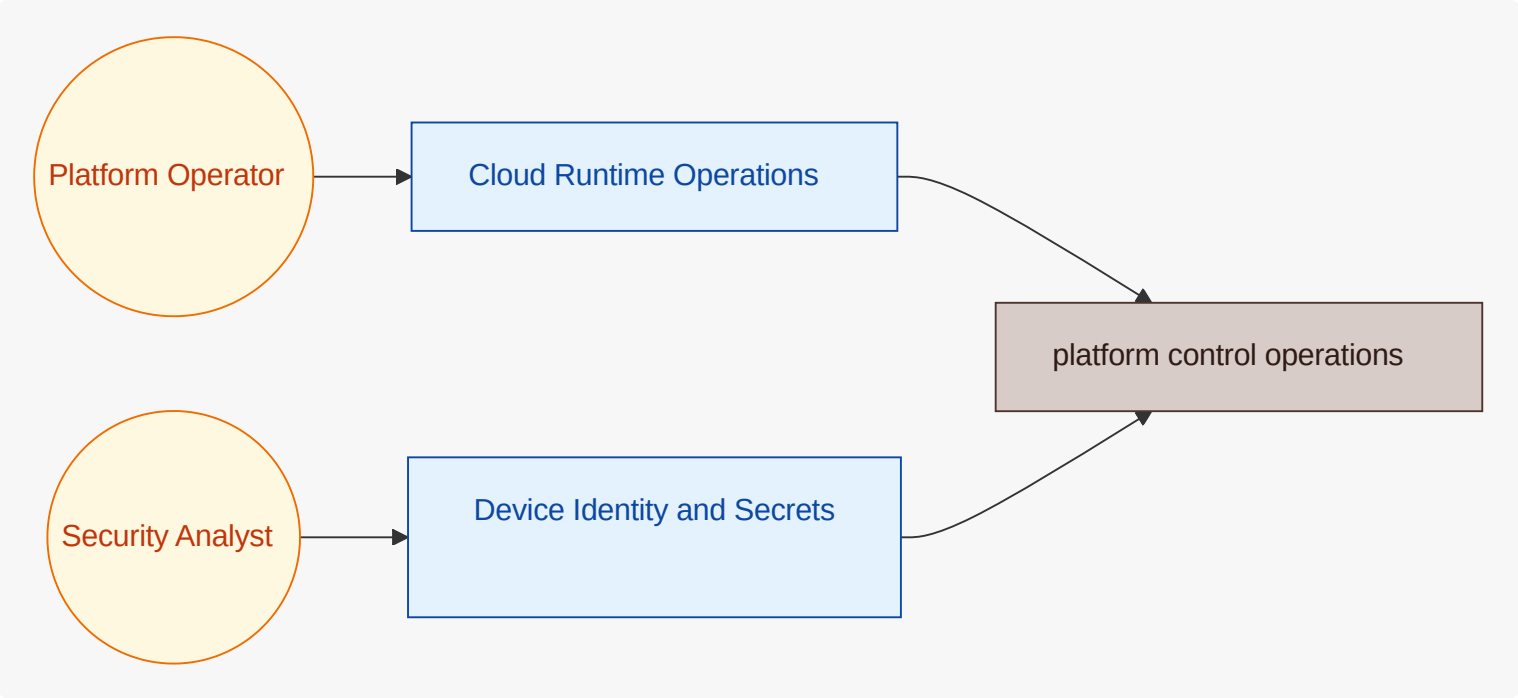



## 7.5. Platform Operations Graph

What this diagram shows: which platform actors operate which platform units and the control-plane targets they influence.

NOTE

Platform operations links are inferred from authored actor mappings plus discovered deployment control-plane artifacts.




## 7.6. Functional Groups and Units (View Scoped)

### 7.6.1. Machine Edge (FG-MACHINE-EDGE)

Machine-edge software artifacts are rolled independently from cloud control to isolate device risk.

### Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

#### Machine Telemetry Collection Deployment

Deployed as edge software component with independent release cadence.

| Deployment Aspect | View-Scoped Detail |
|---|---|
| Deployment trigger context | REQ-COF-001, REQ-COF-006 |
| Deployment dependencies | Device Identity and Secrets, Fleet Ingestion API, IoT Ingest Endpoint, MQTT Device SDK |
| Inferred deployment evidence | runtime: telemetry_retry_queue |

### OTA Update Agent (FU-OTA-UPDATE-AGENT)

#### OTA Update Agent Deployment

Versioned and rolled independently from telemetry pipeline components.

| Deployment Aspect | View-Scoped Detail |
|---|---|
| Deployment trigger context | REQ-COF-003, REQ-COF-004, REQ-COF-005 |
| Deployment dependencies | Device Identity and Secrets, Firmware Bundle Store, Update Campaign Orchestration |
| Inferred deployment evidence | runtime: ota_campaign_schedule, ota_campaign_target |

### 7.6.2. Cloud Control (FG-CLOUD-CONTROL)

Cloud-control units deploy independently to reduce blast radius during campaign logic changes.

#### Fleet Ingestion API (FU-FLEET-INGESTION-API)

##### Fleet Ingestion API Deployment

Deployed as independent cloud ingestion boundary to isolate scaling concerns.

| Deployment Aspect | View-Scoped Detail |
|---|---|
| Deployment trigger context | REQ-COF-001, REQ-COF-002, REQ-COF-006, REQ-COF-008 |
| Deployment dependencies | Cloud Runtime SDK, Device Identity and Secrets, Fleet Observability Reporting |
| Inferred deployment evidence | runtime: fleet_event_topic, fleet_ingestion_api |

#### Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

##### Fleet Observability Reporting Deployment

Deployed as reporting plane with independent scaling and retention controls.

| Deployment Aspect | View-Scoped Detail |
|---|---|
| Deployment trigger context | REQ-COF-002, REQ-COF-005, REQ-COF-007, REQ-COF-008 |
| Deployment dependencies | Cloud Logging Service, Device Identity and Secrets |
| Inferred deployment evidence | runtime: fleet_observability_reporting |

#### Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

##### Update Campaign Orchestration Deployment

Isolated deployment supports safe OTA policy evolution.

| Deployment Aspect | View-Scoped Detail |
|---|---|
| Deployment trigger context | REQ-COF-003, REQ-COF-005 |

| Deployment Aspect | View-Scoped Detail |
|---|---|
| Deployment dependencies | Device Identity and Secrets, Firmware Signer Service, Fleet Observability Reporting |
| Inferred deployment evidence | runtime: update_campaign_orchestration |

### 7.6.3. Platform Operations (FG-PLATFORM-OPERATIONS)

Platform operations governs release sequencing, runtime placement, and operational guardrails.

#### Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

##### Cloud Runtime Operations Deployment

Owns source-to-runtime release controls and rollback guardrails.

| Deployment Aspect | View-Scoped Detail |
|---|---|
| Deployment trigger context | REQ-COF-008 |
| Deployment dependencies | Cloud Logging Service, Device Identity and Secrets |
| Inferred deployment evidence | code modules: cloud_runtime_ops |

#### Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

##### Device Identity and Secrets Deployment

Managed as shared control-plane boundary with controlled key rotation.

| Deployment Aspect | View-Scoped Detail |
|---|---|
| Deployment trigger context | REQ-COF-004, REQ-COF-007 |
| Deployment dependencies | none |
| Inferred deployment evidence | code modules: device_identity_secrets |

# 8. Security View

Show inferred exposure and dependency risk points aligned to unit boundaries, focused on attack paths and security evidence.

## 8.1. What This View Answers

- What are the main trust boundaries and attack paths?
- Which units are exposed and what security evidence exists?
- Where is the threat model incomplete?

## 8.2. Coverage Gaps

Coverage summary: 6/7 units have direct evidence coverage in this view.

- Cloud Runtime Operations has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.

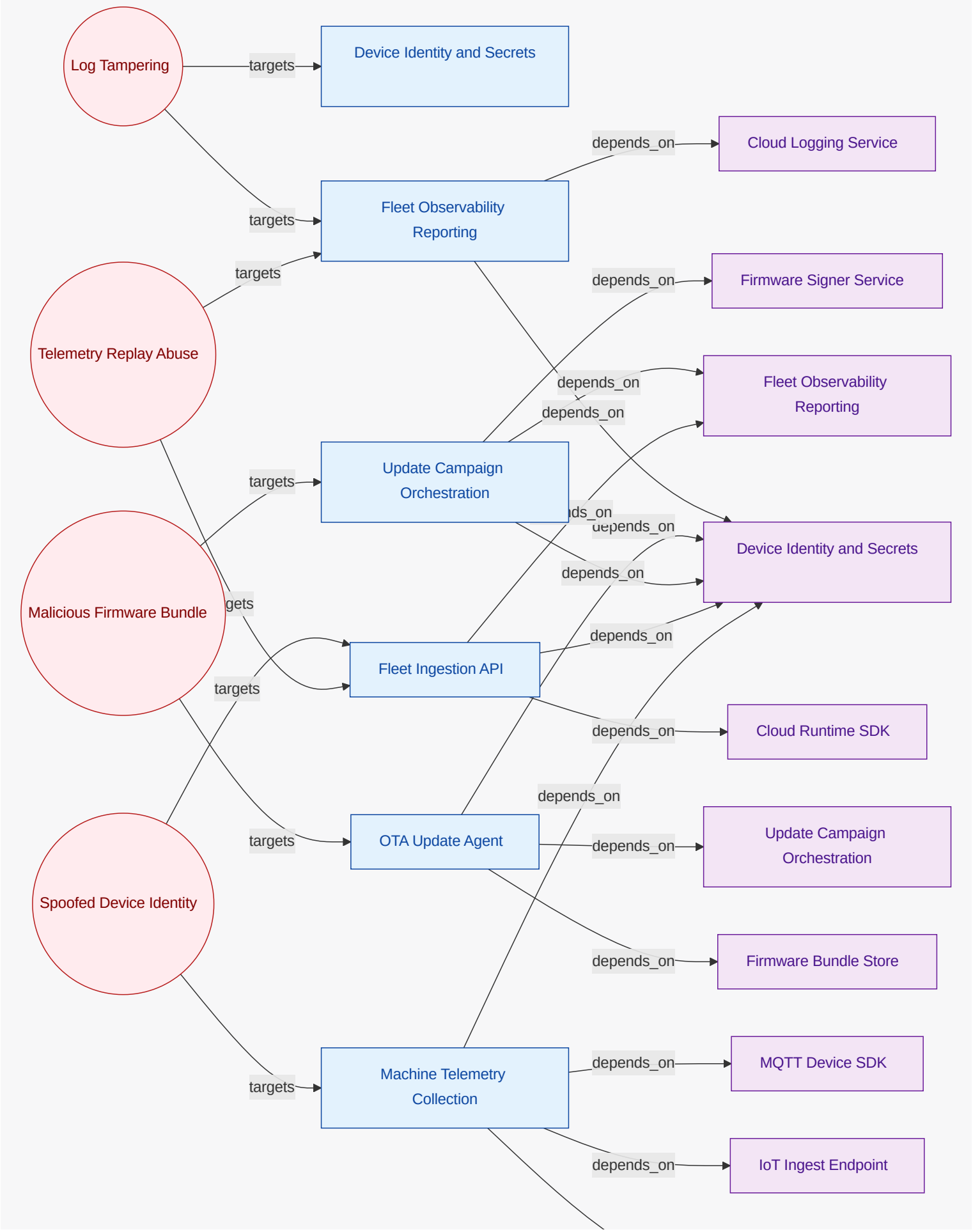
## 8.3. Recommended Next Evidence Additions

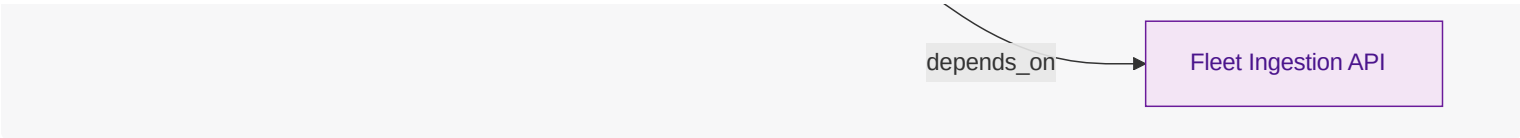
- Add explicit authored attack-vector mappings for units currently marked without threat linkage.
- Add security signal ownership metadata (logs/alerts/events) for unresolved observability evidence.
- Map additional security-relevant dependencies and exposures into authored mappings for traceable coverage.

## 8.4. Attack Path Diagram

What this diagram shows: authored attack vectors, their targeted units, and security-relevant dependency paths.

| | |
|---|---|
| NOTE | Security relationships shown here are explicitly modeled or inferred from available evidence and may be incomplete. Absence of a path does not imply absence of risk. Threats that are currently mitigated (including fraud/abuse threats) should still be modeled as attack vectors for auditability and control-trace visibility. |


Log Tampering
Telemetry Replay Abuse
Malicious Firmware Bundle
Spoofed Device Identity
targets
Device Identity and Secrets
Fleet Observability Reporting
Update Campaign Orchestration
Fleet Ingestion API
OTA Update Agent
Machine Telemetry Collection
depends_on
Cloud Logging Service
Firmware Signer Service
Fleet Observability Reporting
Device Identity and Secrets
Cloud Runtime SDK
Update Campaign Orchestration
Firmware Bundle Store
MQTT Device SDK
IoT Ingest Endpoint




## 8.5. Functional Groups and Units (View Scoped)

### 8.5.1. Machine Edge (FG-MACHINE-EDGE)

Machine Edge enforces firmware trust and authenticated telemetry at the device boundary.

#### Machine Telemetry Collection (FU-MACHINE-TELEMETRY-COLLECTION)

Protects telemetry authenticity and defends against replay abuse patterns.

##### Attack Surface

Device Identity and Secrets, Fleet Ingestion API, IoT Ingest Endpoint, MQTT Device SDK

##### Main Threats

Spoofed Device Identity

##### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

##### Security Evidence Present

runtime: telemetry_retry_queue | code modules: machine_telemetry

##### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

#### OTA Update Agent (FU-OTA-UPDATE-AGENT)

Enforces signature validation and rejects untrusted firmware bundles.

##### Attack Surface

Device Identity and Secrets, Firmware Bundle Store, Update Campaign Orchestration

##### Main Threats

Malicious Firmware Bundle

##### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

##### Security Evidence Present

runtime: ota_campaign_schedule, ota_campaign_target | code modules: ota_update_agent

##### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

### 8.5.2. Cloud Control (FG-CLOUD-CONTROL)

Cloud Control focuses on replay detection, campaign authorization, and audit integrity.

#### Fleet Ingestion API (FU-FLEET-INGESTION-API)

Applies identity checks and replay throttling for ingest requests.

#### Attack Surface

Cloud Runtime SDK, Device Identity and Secrets, Fleet Observability Reporting

#### Main Threats

Spoofed Device Identity, Telemetry Replay Abuse

#### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

#### Security Evidence Present

runtime: fleet_event_topic, fleet_ingestion_api | code modules: fleet_ingestion_api

#### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

### Fleet Observability Reporting (FU-FLEET-OBSERVABILITY-REPORTING)

Preserves audit integrity and protects sensitive identifiers in persisted records.

#### Attack Surface

Cloud Logging Service, Device Identity and Secrets

#### Main Threats

Log Tampering, Telemetry Replay Abuse

#### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

#### Security Evidence Present

runtime: fleet_observability_reporting | code modules: fleet_observability_reporting

#### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

### Update Campaign Orchestration (FU-UPDATE-CAMPAIGN-ORCHESTRATION)

Enforces signed-firmware policy and campaign authorization boundaries.

#### Attack Surface

Device Identity and Secrets, Firmware Signer Service, Fleet Observability Reporting

#### Main Threats

Malicious Firmware Bundle

#### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

#### Security Evidence Present

runtime: update_campaign_orchestration | code modules: update_campaign_orchestrator

#### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

### 8.5.3. Platform Operations (FG-PLATFORM-OPERATIONS)

Platform operations hardens identity, key rotation, and log integrity boundaries.

#### Cloud Runtime Operations (FU-CLOUD-RUNTIME-OPERATIONS)

Enforces operational hardening and security escalation signaling.

#### Attack Surface

Cloud Logging Service, Device Identity and Secrets

#### Main Threats

no explicit attack vector targeting this unit

#### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

#### Security Evidence Present

code modules: cloud_runtime_ops

#### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

#### Device Identity and Secrets (FU-DEVICE-IDENTITY-SECRETS)

Protects credential lifecycle and enforces least-privilege access patterns.

#### Attack Surface

none

#### Main Threats

Log Tampering

#### Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

#### Security Evidence Present

code modules: device_identity_secrets

#### Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

# 9. Traceability View

Show requirement-to-unit-to-evidence traceability, including coverage confidence and explicit evidence gaps.

## 9.1. What This View Answers

- Which requirements map to which units?
- Which authored units have runtime/code evidence and which do not?
- Where are confidence or completeness gaps in trace coverage?

## 9.2. Coverage Gaps

Coverage summary: 7/7 units have direct evidence coverage in this view.

- Cloud Runtime Operations has no direct inferred runtime evidence yet.
- Device Identity and Secrets has no direct inferred runtime evidence yet.

## 9.3. Recommended Next Evidence Additions

- Add explicit requirement-to-owner mappings where requirement scope is currently broad or ambiguous.
- Add ownership annotations for runtime and code artifacts where coverage is still unresolved.
- Prioritize closing gaps for units that are authored but lack both runtime and code evidence.

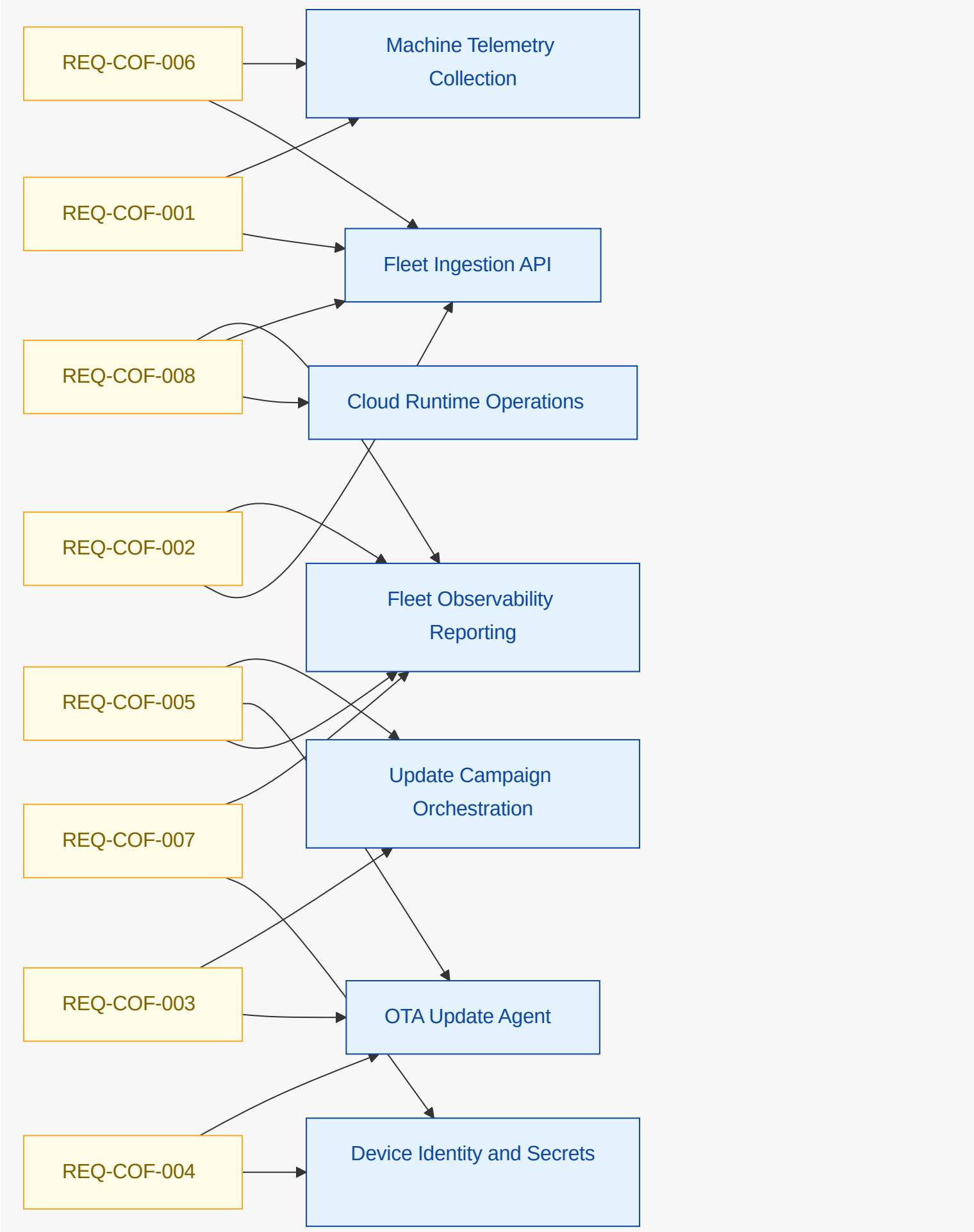
## 9.4. Traceability Evidence Graph

What this diagram shows: authored units and the inferred evidence links that support traceability confidence.

> **NOTE** This graph summarizes inferred ownership/evidence links and is intended for traceability confidence checks.

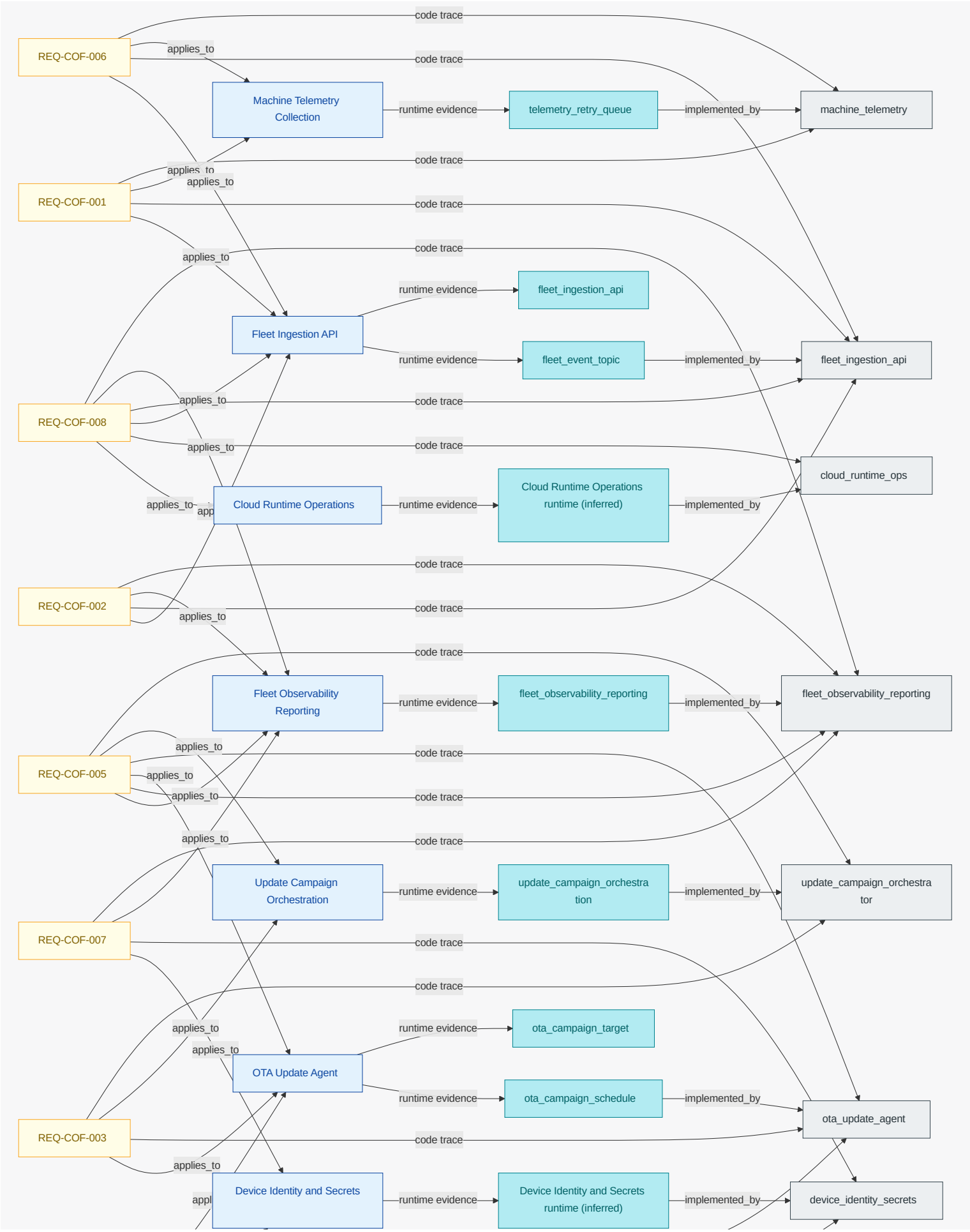
## 9.5. Requirement-to-Unit Mapping

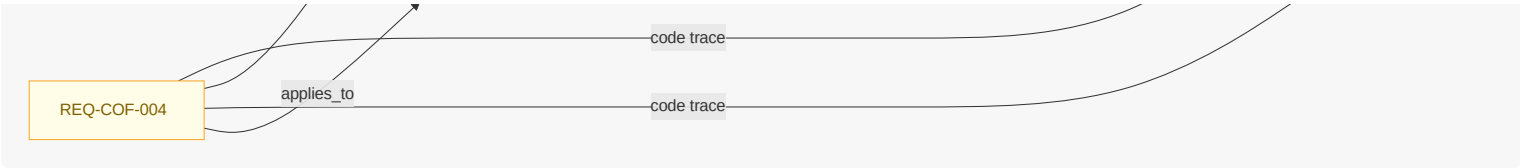
What this diagram shows: direct authored mapping from requirements to the functional units they apply to.


REQ-COF-006
Machine Telemetry Collection
REQ-COF-001
Fleet Ingestion API
REQ-COF-008
Cloud Runtime Operations
REQ-COF-002
Fleet Observability Reporting
REQ-COF-005
Update Campaign Orchestration
REQ-COF-007
REQ-COF-003
OTA Update Agent
REQ-COF-004
Device Identity and Secrets

## 9.6. Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: requirement-to-unit links extended with inferred runtime/code evidence attached to each unit.


REQ-COF-006
REQ-COF-001
REQ-COF-008
REQ-COF-002
REQ-COF-005
REQ-COF-007
REQ-COF-003
Machine Telemetry Collection
Fleet Ingestion API
Cloud Runtime Operations
Fleet Observability Reporting
Update Campaign Orchestration
OTA Update Agent
Device Identity and Secrets
telemetry_retry_queue
fleet_ingestion_api
fleet_event_topic
Cloud Runtime Operations runtime (inferred)
fleet_observability_reporting
update_campaign_orchestration
ota_campaign_target
ota_campaign_schedule
Device Identity and Secrets runtime (inferred)
machine_telemetry
fleet_ingestion_api
cloud_runtime_ops
fleet_observability_reporting
update_campaign_orchestrator
ota_update_agent
device_identity_secrets
applies_to
code trace
runtime evidence
implemented_by




Show requirement-to-unit mappings inferred from appliesTo and authored architecture ownership boundaries.

# 10. Traceability Appendix

## 10.1. Requirement Details

### 10.1.1. REQ-COF-001

While coffee machine fleet operation is active, when brew cycle completed event is received, the coffee fleet system shall publish brew telemetry to fleet ingestion api. Notes: Primary machine-to-cloud telemetry contract.

### 10.1.2. REQ-COF-002

While coffee machine fleet operation is active, when telemetry ingested event is received, the coffee fleet system shall persist fleet telemetry metrics and logging records for fleet operator. Notes: Operational observability requirement.

### 10.1.3. REQ-COF-003

Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall distribute signed firmware bundle to targeted machine cohort. Notes: OTA rollout behavior from central cloud control.

### 10.1.4. REQ-COF-004

Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall gate firmware apply on firmware signature is valid. Notes: Firmware integrity and trust boundary control.

### 10.1.5. REQ-COF-005

If firmware validation failed event is received, then the coffee fleet system shall rollback firmware apply and report ota status to update campaign orchestration. Notes: Rollback and failure signaling path.

### 10.1.6. REQ-COF-006

If cloud ingest endpoint is unavailable, then the coffee fleet system shall queue telemetry for retry and flush when normal mode is active. Notes: Resilience for temporary cloud outage.

### 10.1.7. REQ-COF-007

Where audit logging is enabled, when ota status reported event is received, the coffee fleet system shall persist redacted audit log record including machine identity and campaign id. Notes: Audit/compliance logging requirement.

### 10.1.8. REQ-COF-008

If telemetry replay abuse is detected, then the coffee fleet system shall throttle duplicate telemetry and notify security analyst. Notes: Abuse control and security response requirement.

## 10.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

### 10.2.1. Ota Rollout Flow

| Field | Value |
|---|---|
| ID | VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC |

| Field | Value |
|---|---|
| Kind | e2e |
| Status | partial |
| Verifies Requirements | REQ-COF-003, REQ-COF-005 |
| Test Code Elements | tests/e2e/ota_rollout_flow.yaml |
| Derived Functional Owners | FU-FLEET-OBSERVABILITY-REPORTING, FU-OTA-UPDATE-AGENT, FU-UPDATE-CAMPAIGN-ORCHESTRATION |
| Result Summary | not-run:1, partial:1 |
| Evidence | test-results/ota-rollback-report.md, tests/e2e/ota_rollout_flow.yaml |
| Notes | Inferred from test source artifact. |

### 10.2.2. Logging Pipeline Test

| Field | Value |
|---|---|
| ID | VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF |
| Kind | integration |
| Status | not-run |
| Verifies Requirements | REQ-COF-002, REQ-COF-007 |
| Test Code Elements | tests/integration/logging_pipeline_test.ts |
| Derived Functional Owners | FU-DEVICE-IDENTITY-SECRETS, FU-FLEET-INGESTION-API, FU-FLEET-OBSERVABILITY-REPORTING |
| Result Summary | not-run:1, pass:1 |
| Evidence | test-results/integration-logging.json, tests/integration/logging_pipeline_test.ts |
| Notes | Inferred from test source artifact. |

### 10.2.3. Telemetry Ingest Contract Test

| Field | Value |
|---|---|
| ID | VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794 |
| Kind | test |
| Status | pass |

| Field | Value |
|---|---|
| Verifies Requirements | REQ-COF-001 |
| Test Code Elements | tests/contract/telemetry_ingest_contract_test.go |
| Derived Functional Owners | FU-FLEET-INGESTION-API, FU-MACHINE-TELEMETRY-COLLECTION |
| Result Summary | pass:1 |
| Evidence | test-results/contract-telemetry.json, tests/contract/telemetry_ingest_contract_test.go |
| Notes | Inferred from test source artifact. |

### 10.2.4. Ota Campaign Test

| Field | Value |
|---|---|
| ID | VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8 |
| Kind | unit |
| Status | pass |
| Verifies Requirements | REQ-COF-003, REQ-COF-004 |
| Test Code Elements | tests/unit/ota_campaign_test.go |
| Derived Functional Owners | FU-DEVICE-IDENTITY-SECRETS, FU-OTA-UPDATE-AGENT, FU-UPDATE-CAMPAIGN-ORCHESTRATION |
| Result Summary | pass:2 |
| Evidence | test-results/unit-ota.xml, tests/unit/ota_campaign_test.go |
| Notes | Inferred from test source artifact. |

## 10.3. Verification Result Mapping

What this section shows: per-requirement result mapping inferred from test-result artifacts.

| Check ID | Requirement | Result | Evidence / Notes |
|---|---|---|---|
| VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794 | REQ-COF-001 | pass | test-results/contract-telemetry.json<br>ingest payload accepted with machine identity and telemetry fields |

| Check ID | Requirement | Result | Evidence / Notes |
|---|---|---|---|
| VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF | REQ-COF-002 | not-run | none<br>No parsed test-result artifact for this requirement yet. |
| VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC | REQ-COF-003 | not-run | none<br>No parsed test-result artifact for this requirement yet. |
| VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8 | REQ-COF-003 | pass | test-results/unit-ota.xml |
| VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8 | REQ-COF-004 | pass | test-results/unit-ota.xml |
| VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC | REQ-COF-005 | partial | test-results/ota-rollback-report.md |
| VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF | REQ-COF-007 | pass | test-results/integration-logging.json<br>audit log record persisted with redacted machine identity |

# 11. Reference Index

## 11.1. Authored References

### ACT-BARISTA

Kind: Actor Name: Barista

Operates coffee machines and observes local device status.

### ACT-FLEET-OPERATOR

Kind: Actor Name: Fleet Operator

Manages OTA campaigns and fleet-wide operational decisions.

### ACT-PLATFORM-OPERATOR

Kind: Actor Name: Platform Operator

Operates cloud runtime, deployment, and reliability controls.

### ACT-SECURITY-ANALYST

Kind: Actor Name: Security Analyst

Reviews firmware integrity, identity controls, and audit logging.

### AV-LOG-TAMPERING

Kind: Attack Vector Name: Log Tampering

Attempts to suppress or alter audit/operational logs.

### AV-MALICIOUS-FIRMWARE-BUNDLE

Kind: Attack Vector Name: Malicious Firmware Bundle

Tampered OTA payload targeting update agent execution.

### AV-SPOOFED-DEVICE-IDENTITY

Kind: Attack Vector Name: Spoofed Device Identity

Unauthorized identity attempting telemetry/command access.

### AV-TELEMETRY-REPLAY-ABUSE

Kind: Attack Vector Name: Telemetry Replay Abuse

Replayed telemetry intended to overload ingestion and corrupt insights.

### FG-CLOUD-CONTROL

Kind: Functional Group Name: Cloud Control

Central cloud ingestion, campaign control, and fleet reporting.

### FG-MACHINE-EDGE

Kind: Functional Group Name: Machine Edge

Device-local telemetry and OTA execution responsibilities.

### FG-PLATFORM-OPERATIONS

Kind: Functional Group Name: Platform Operations

Cloud runtime operations plus identity and secret-management controls.

### FU-CLOUD-RUNTIME-OPERATIONS

Kind: Functional Unit Name: Cloud Runtime Operations

Owns runtime deployment lifecycle, scaling controls, and runtime operational safety.

### FU-DEVICE-IDENTITY-SECRETS

Kind: Functional Unit Name: Device Identity and Secrets

Manages device credentials, cloud tokens, and secure configuration used by edge/cloud units.

### FU-FLEET-INGESTION-API

Kind: Functional Unit Name: Fleet Ingestion API

Receives telemetry and machine status, applies validation, and emits ingestion events.

### FU-FLEET-OBSERVABILITY-REPORTING

Kind: Functional Unit Name: Fleet Observability Reporting

Aggregates telemetry and OTA outcomes into dashboards, alerts, and audit records.

### FU-MACHINE-TELEMETRY-COLLECTION

Kind: Functional Unit Name: Machine Telemetry Collection

Collects brew/device telemetry and transmits cloud-bound telemetry payloads.

### FU-OTA-UPDATE-AGENT

Kind: Functional Unit Name: OTA Update Agent

Downloads signed firmware bundles, validates integrity, applies updates, and reports status.

### FU-UPDATE-CAMPAIGN-ORCHESTRATION

Kind: Functional Unit Name: Update Campaign Orchestration

Plans OTA rollout cohorts, drives update commands, and handles rollback flow decisions.

### REF-CLOUD-LOGGING-SERVICE

Kind: Referenced Element Name: Cloud Logging Service

platform_service / runtime

### REF-CLOUD-RUNTIME-SDK

Kind: Referenced Element Name: Cloud Runtime SDK

third_party_library / code

### REF-FIRMWARE-BUNDLE-STORE

Kind: Referenced Element Name: Firmware Bundle Store

object_store / runtime

### REF-FIRMWARE-SIGNER-SERVICE

Kind: Referenced Element Name: Firmware Signer Service

signing_service / runtime

### REF-IOT-INGEST-ENDPOINT

Kind: Referenced Element Name: IoT Ingest Endpoint

external_service_endpoint / runtime

### REF-MQTT-DEVICE-SDK

Kind: Referenced Element Name: MQTT Device SDK

third_party_library / code

### REQ-COF-001

Kind: Requirement Name: REQ-COF-001 Resolves to: canonical While coffee machine fleet operation is active, when brew cycle completed event is received, the coffee fleet system shall publish brew telemetry to fleet ingestion api.

### REQ-COF-002

Kind: Requirement Name: REQ-COF-002 Resolves to: canonical While coffee machine fleet operation is active, when telemetry ingested event is received, the coffee fleet system shall persist fleet telemetry metrics and logging records for fleet operator.

### REQ-COF-003

Kind: Requirement Name: REQ-COF-003 Resolves to: canonical Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall distribute signed firmware bundle to targeted machine cohort.

### REQ-COF-004

Kind: Requirement Name: REQ-COF-004 Resolves to: canonical Where ota campaign is enabled, when ota campaign scheduled event is received, the coffee fleet system shall gate firmware apply on firmware signature is valid.

### REQ-COF-005

Kind: Requirement Name: REQ-COF-005 Resolves to: canonical If firmware validation failed event is received, then the coffee fleet system shall rollback firmware apply and report ota status to update campaign orchestration.

### REQ-COF-006

Kind: Requirement Name: REQ-COF-006 Resolves to: canonical If cloud ingest endpoint is unavailable, then the coffee fleet system shall queue telemetry for retry and flush when normal mode is active.

### REQ-COF-007

Kind: Requirement Name: REQ-COF-007 Resolves to: canonical Where audit logging is enabled, when ota status reported event is received, the coffee fleet system shall persist redacted audit log record including machine identity and campaign id.

### REQ-COF-008

Kind: Requirement Name: REQ-COF-008 Resolves to: canonical If telemetry replay abuse is detected, then the coffee fleet system shall throttle duplicate telemetry and notify security analyst.

## 11.2. Catalog References

### ACT-BARISTA

Kind: Catalog Actor Name: barista Resolves to: canonical Operates coffee machines and observes local machine health.

### ACT-FLEET-OPERATOR

Kind: Catalog Actor Name: fleet operator Resolves to: canonical Owns OTA rollout campaigns and fleet-level reliability.

### ACT-PLATFORM-OPERATOR

Kind: Catalog Actor Name: platform operator Resolves to: canonical Owns runtime deployment and operational controls.

### ACT-SECURITY-ANALYST

Kind: Catalog Actor Name: security analyst Resolves to: canonical Reviews identity, firmware integrity, and logging controls.

### audit enabled

Kind: Catalog Alias Name: audit logging is enabled Resolves to: canonical Alias of FEAT-AUDIT-LOGGING

### audit tampering

Kind: Catalog Alias Name: log tampering Resolves to: canonical Alias of AV-LOG-TAMPERING

### brew completed

Kind: Catalog Alias Name: brew cycle completed event is received Resolves to: canonical Alias of EVT-BREW-CYCLE-COMPLETED

### campaign orchestration

Kind: Catalog Alias Name: update campaign orchestration Resolves to: canonical Alias of FU-UPDATE-CAMPAIGN-ORCHESTRATION

### campaign scheduled

Kind: Catalog Alias Name: ota campaign scheduled event is received Resolves to: canonical Alias of EVT-OTA-CAMPAIGN-SCHEDULED

### central cloud site

Kind: Catalog Alias Name: cloud control Resolves to: canonical Alias of FG-CLOUD-CONTROL

### cloud sdk

Kind: Catalog Alias Name: cloud runtime sdk Resolves to: canonical Alias of REF-CLOUD-RUNTIME-SDK

### coffee machine edge

Kind: Catalog Alias Name: machine edge Resolves to: canonical Alias of FG-MACHINE-EDGE

### device secrets

Kind: Catalog Alias Name: device identity and secrets Resolves to: canonical Alias of FU-DEVICE-IDENTITY-SECRETS

### device spoofing

Kind: Catalog Alias Name: spoofed device identity Resolves to: canonical Alias of AV-SPOOFED-DEVICE-IDENTITY

### device update agent

Kind: Catalog Alias Name: ota update agent Resolves to: canonical Alias of FU-OTA-UPDATE-AGENT

### firmware store

Kind: Catalog Alias Name: firmware bundle store Resolves to: canonical Alias of REF-FIRMWARE-BUNDLE-STORE

### firmware tampering

Kind: Catalog Alias Name: malicious firmware bundle Resolves to: canonical Alias of AV-MALICIOUS-FIRMWARE-BUNDLE

### fleet admin

Kind: Catalog Alias Name: fleet operator Resolves to: canonical Alias of ACT-FLEET-OPERATOR

### fleet operation active

Kind: Catalog Alias Name: coffee machine fleet operation is active Resolves to: canonical Alias of STATE-FLEET-OPERATION-ACTIVE

### fleet platform operations

Kind: Catalog Alias Name: platform operations Resolves to: canonical Alias of FG-PLATFORM-OPERATIONS

### ingestion api

Kind: Catalog Alias Name: fleet ingestion api Resolves to: canonical Alias of FU-FLEET-INGESTION-API

### iot endpoint

Kind: Catalog Alias Name: iot ingest endpoint Resolves to: canonical Alias of REF-IOT-INGEST-ENDPOINT

### logging service

Kind: Catalog Alias Name: cloud logging service Resolves to: canonical Alias of REF-CLOUD-LOGGING-SERVICE

### mqtt sdk

Kind: Catalog Alias Name: mqtt device sdk Resolves to: canonical Alias of REF-MQTT-DEVICE-SDK

### normal mode

Kind: Catalog Alias Name: normal mode is active Resolves to: canonical Alias of MODE-NORMAL

### observability reporting

Kind: Catalog Alias Name: fleet observability reporting Resolves to: canonical Alias of FU-FLEET-OBSERVABILITY-REPORTING

### ota enabled

Kind: Catalog Alias Name: ota campaign is enabled Resolves to: canonical Alias of FEAT-OTA-CAMPAIGN

### ota status reported

Kind: Catalog Alias Name: ota status reported event is received Resolves to: canonical Alias of EVT-OTA-STATUS-REPORTED

### platform ops

Kind: Catalog Alias Name: platform operator Resolves to: canonical Alias of ACT-PLATFORM-OPERATOR

### runtime operations

Kind: Catalog Alias Name: cloud runtime operations Resolves to: canonical Alias of FU-CLOUD-RUNTIME-OPERATIONS

### security reviewer

Kind: Catalog Alias Name: security analyst Resolves to: canonical Alias of ACT-SECURITY-ANALYST

### signature valid

Kind: Catalog Alias Name: firmware signature is valid Resolves to: canonical Alias of COND-FIRMWARE-SIGNATURE-VALID

### signer service

Kind: Catalog Alias Name: firmware signer service Resolves to: canonical Alias of REF-FIRMWARE-SIGNER-SERVICE

### store operator

Kind: Catalog Alias Name: barista Resolves to: canonical Alias of ACT-BARISTA

### telemetry collection

Kind: Catalog Alias Name: machine telemetry collection Resolves to: canonical Alias of FU-MACHINE-TELEMETRY-COLLECTION

### telemetry ingested

Kind: Catalog Alias Name: telemetry ingested event is received Resolves to: canonical Alias of EVT-TELEMETRY-INGESTED

### telemetry replay

Kind: Catalog Alias Name: telemetry replay abuse Resolves to: canonical Alias of AV-TELEMETRY-REPLAY-ABUSE

### validation failed

Kind: Catalog Alias Name: firmware validation failed event is received Resolves to: canonical Alias of EVT-FIRMWARE-VALIDATION-FAILED

### AV-LOG-TAMPERING

Kind: Catalog Attack Vector Name: log tampering Resolves to: canonical Attempts to suppress or alter security/operational log records.

### AV-MALICIOUS-FIRMWARE-BUNDLE

Kind: Catalog Attack Vector Name: malicious firmware bundle Resolves to: canonical Tampered firmware payload attempts to execute on devices.

### AV-SPOOFED-DEVICE-IDENTITY

Kind: Catalog Attack Vector Name: spoofed device identity Resolves to: canonical Unauthorized actor impersonates a machine identity.

### AV-TELEMETRY-REPLAY-ABUSE

Kind: Catalog Attack Vector Name: telemetry replay abuse Resolves to: canonical Replayed telemetry floods ingestion and distorts reporting.

### COND-FIRMWARE-SIGNATURE-VALID

Kind: Catalog Condition Name: firmware signature is valid Resolves to: canonical Condition indicating firmware signature verification succeeded.

### DATA-AUDIT-LOG-RECORD

Kind: Catalog Data Term Name: audit log record Resolves to: canonical Immutable log record for security and compliance analysis.

### DATA-BREW-TELEMETRY

Kind: Catalog Data Term Name: brew telemetry Resolves to: canonical Machine telemetry including brew duration, temperature, and status metadata.

### DATA-OTA-CAMPAIGN-POLICY

Kind: Catalog Data Term Name: ota campaign policy Resolves to: canonical Policy describing rollout cohorts, windows, and rollback criteria.

### DATA-OTA-STATUS

Kind: Catalog Data Term Name: ota status Resolves to: canonical Device-level status for firmware download, apply, and rollback outcomes.

### EVT-BREW-CYCLE-COMPLETED

Kind: Catalog Event Name: brew cycle completed event is received Resolves to: canonical Event raised when a coffee machine completes a brew cycle.

### EVT-FIRMWARE-VALIDATION-FAILED

Kind: Catalog Event Name: firmware validation failed event is received Resolves to: canonical Event raised when signature/hash validation fails on firmware.

### EVT-OTA-CAMPAIGN-SCHEDULED

Kind: Catalog Event Name: ota campaign scheduled event is received Resolves to: canonical Event raised when a fleet OTA campaign is scheduled.

### EVT-OTA-STATUS-REPORTED

Kind: Catalog Event Name: ota status reported event is received Resolves to: canonical Event raised when device reports OTA apply status.

### EVT-TELEMETRY-INGESTED

Kind: Catalog Event Name: telemetry ingested event is received Resolves to: canonical Event raised when cloud ingestion accepts telemetry.

### FEAT-AUDIT-LOGGING

Kind: Catalog Feature Name: audit logging is enabled Resolves to: canonical Feature flag enabling audit-grade log persistence.

### FEAT-OTA-CAMPAIGN

Kind: Catalog Feature Name: ota campaign is enabled Resolves to: canonical Feature flag enabling centralized OTA rollout behavior.

### FG-CLOUD-CONTROL

Kind: Catalog Functional Group Name: cloud control Resolves to: canonical Capability area for fleet ingestion, campaign orchestration, and cloud reporting.

### FG-MACHINE-EDGE

Kind: Catalog Functional Group Name: machine edge Resolves to: canonical Capability area for device-local telemetry and OTA execution behavior.

### FG-PLATFORM-OPERATIONS

Kind: Catalog Functional Group Name: platform operations Resolves to: canonical Capability area for runtime operations and device identity/secrets lifecycle.

### FU-CLOUD-RUNTIME-OPERATIONS

Kind: Catalog Functional Unit Name: cloud runtime operations Resolves to: canonical Operates cloud runtime deployment and scaling controls.

### FU-DEVICE-IDENTITY-SECRETS

Kind: Catalog Functional Unit Name: device identity and secrets Resolves to: canonical Manages device identity material and secure configuration.

### FU-FLEET-INGESTION-API

Kind: Catalog Functional Unit Name: fleet ingestion api Resolves to: canonical Ingests telemetry and machine status from edge devices.

### FU-FLEET-OBSERVABILITY-REPORTING

Kind: Catalog Functional Unit Name: fleet observability reporting Resolves to: canonical Aggregates telemetry and update outcomes into operational and audit signals.

### FU-MACHINE-TELEMETRY-COLLECTION

Kind: Catalog Functional Unit Name: machine telemetry collection Resolves to: canonical Collects brew/device telemetry and publishes cloud-bound payloads.

### FU-OTA-UPDATE-AGENT

Kind: Catalog Functional Unit Name: ota update agent Resolves to: canonical Downloads, validates, and applies signed firmware updates on machines.

### FU-UPDATE-CAMPAIGN-ORCHESTRATION

Kind: Catalog Functional Unit Name: update campaign orchestration Resolves to: canonical Selects target machine groups and coordinates OTA rollout phases.

### MODE-NORMAL

Kind: Catalog Mode Name: normal mode is active Resolves to: canonical Default operation mode for telemetry and OTA behavior.

### REF-CLOUD-LOGGING-SERVICE

Kind: Catalog Referenced Element Name: cloud logging service Resolves to: canonical Managed log and metrics service used for fleet observability.

### REF-CLOUD-RUNTIME-SDK

Kind: Catalog Referenced Element Name: cloud runtime sdk Resolves to: canonical SDK used by cloud units for runtime integrations.

### REF-FIRMWARE-BUNDLE-STORE

Kind: Catalog Referenced Element Name: firmware bundle store Resolves to: canonical Object store containing signed OTA firmware bundles.

### REF-FIRMWARE-SIGNER-SERVICE

Kind: Catalog Referenced Element Name: firmware signer service Resolves to: canonical Service that signs firmware artifacts before OTA rollout.

### REF-IOT-INGEST-ENDPOINT

Kind: Catalog Referenced Element Name: iot ingest endpoint Resolves to: canonical Cloud endpoint for receiving telemetry and status from devices.

### REF-MQTT-DEVICE-SDK

Kind: Catalog Referenced Element Name: mqtt device sdk Resolves to: canonical SDK used by edge components for telemetry and command transport.

### STATE-FLEET-OPERATION-ACTIVE

Kind: Catalog State Name: coffee machine fleet operation is active Resolves to: canonical State where machines and central cloud are actively serving operations.

## 11.3. Inferred Runtime References

### ota_campaign_schedule

Kind: Runtime eventbridge_rule Owner: FU-OTA-UPDATE-AGENT Source: examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf

### ota_campaign_target

Kind: Runtime eventbridge_target Owner: FU-OTA-UPDATE-AGENT Source: examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf

### fleet_ingestion_api

Kind: Runtime lambda_function Owner: FU-FLEET-INGESTION-API Source: examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf

### fleet_observability_reporting

Kind: Runtime lambda_function Owner: FU-FLEET-OBSERVABILITY-REPORTING Source: examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf

### update_campaign_orchestration

Kind: Runtime lambda_function Owner: FU-UPDATE-CAMPAIGN-ORCHESTRATION Source: examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf

### telemetry_retry_queue

Kind: Runtime queue Owner: FU-MACHINE-TELEMETRY-COLLECTION Source: examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf

### fleet_event_topic

Kind: Runtime topic Owner: FU-FLEET-INGESTION-API Source: examples/coffee-fleet-ota-cloud-sample/infra/terraform/main.tf

## 11.4. Inferred Code References

### cloud_runtime_ops.go

Kind: Code source_file Owner: FU-CLOUD-RUNTIME-OPERATIONS Source: cloud_runtime_ops.go

### device_identity_secrets.ts

Kind: Code source_file Owner: FU-DEVICE-IDENTITY-SECRETS Source: device_identity_secrets.ts

### fleet_ingestion_api.go

Kind: Code source_file Owner: FU-FLEET-INGESTION-API Source: fleet_ingestion_api.go

### fleet_observability_reporting.ts

Kind: Code source_file Owner: FU-FLEET-OBSERVABILITY-REPORTING Source: fleet_observability_reporting.ts

### machine_telemetry.rs

Kind: Code source_file Owner: FU-MACHINE-TELEMETRY-COLLECTION Source: machine_telemetry.rs

### ota_update_agent.rs

Kind: Code source_file Owner: FU-OTA-UPDATE-AGENT Source: ota_update_agent.rs

### update_campaign_orchestrator.ts

Kind: Code source_file Owner: FU-UPDATE-CAMPAIGN-ORCHESTRATION Source: update_campaign_orchestrator.ts

### CODE-HANDLEROLLBACK

Kind: Code symbol Owner: FU-UPDATE-CAMPAIGN-ORCHESTRATION Source: update_campaign_orchestrator.ts:9

### CODE-INGESTTELEMETRY

Kind: Code symbol Owner: FU-FLEET-INGESTION-API Source: fleet_ingestion_api.go:10

### CODE-NOTIFYREPLAYABUSE

Kind: Code symbol Owner: FU-FLEET-OBSERVABILITY-REPORTING Source: fleet_observability_reporting.ts:14

### CODE-PERSISTAUDITLOG

Kind: Code symbol Owner: FU-FLEET-OBSERVABILITY-REPORTING Source: fleet_observability_reporting.ts:9

### CODE-PERSISTTELEMETRYMETRIC

Kind: Code symbol Owner: FU-FLEET-OBSERVABILITY-REPORTING Source: fleet_observability_reporting.ts:4

### CODE-PLANCAMPAIGN

Kind: Code symbol Owner: FU-UPDATE-CAMPAIGN-ORCHESTRATION Source: update_campaign_orchestrator.ts:4

### CODE-PUBLISHBREWTELEMETRY

Kind: Code symbol Owner: FU-MACHINE-TELEMETRY-COLLECTION Source: machine_telemetry.rs:9

### CODE-QUEUEFORRETRY

Kind: Code symbol Owner: FU-MACHINE-TELEMETRY-COLLECTION Source: machine_telemetry.rs:14

### CODE-RAISESECURITYNOTIFICATION

Kind: Code symbol Owner: FU-CLOUD-RUNTIME-OPERATIONS Source: cloud_runtime_ops.go:5

### CODE-REDACTIDENTITYFORAUDIT

Kind: Code symbol Owner: FU-DEVICE-IDENTITY-SECRETS Source: device_identity_secrets.ts:9

### CODE-REJECTREPLAY

Kind: Code symbol Owner: FU-FLEET-INGESTION-API Source: fleet_ingestion_api.go:15

### CODE-ROLLBACKANDREPORT

Kind: Code symbol Owner: FU-OTA-UPDATE-AGENT Source: ota_update_agent.rs:13

### CODE-VALIDATEANDAPPLY

Kind: Code symbol Owner: FU-OTA-UPDATE-AGENT Source: ota_update_agent.rs:4

### CODE-VERIFYFIRMWARESIGNATURE

Kind: Code symbol Owner: FU-DEVICE-IDENTITY-SECRETS Source: device_identity_secrets.ts:4

## 11.5. Verification References

### VER-INF-E2E-OTA_ROLLOUT_FLOW-8CBDAC

Kind: Verification Check Name: Ota Rollout Flow Resolves to: canonical Kind: e2e; Status: partial; Evidence: test-results/ota-rollback-report.md, tests/e2e/ota_rollout_flow.yaml

### VER-INF-INTEGRATION-LOGGING_PIPELINE_TEST-B39BBF

Kind: Verification Check Name: Logging Pipeline Test Resolves to: canonical Kind: integration; Status: not-run; Evidence: test-results/integration-logging.json, tests/integration/logging_pipeline_test.ts

### VER-INF-TEST-TELEMETRY_INGEST_CONTRACT_TEST-394794

Kind: Verification Check Name: Telemetry Ingest Contract Test Resolves to: canonical Kind: test; Status: pass; Evidence: test-results/contract-telemetry.json, tests/contract/telemetry_ingest_contract_test.go

### VER-INF-UNIT-OTA_CAMPAIGN_TEST-5CC0D8

Kind: Verification Check Name: Ota Campaign Test Resolves to: canonical Kind: unit; Status: pass; Evidence: test-results/unit-ota.xml, tests/unit/ota_campaign_test.go